

The Immune System

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Introduction

- This lesson explains how the immune system defends the body using both nonspecific and specific immunity.
- It highlights the roles of barriers, white blood cells, antibodies, and immune memory.

1. Nonspecific Immunity

- a. First line of defense that targets all pathogens, not specific ones.
- b. Barriers:
 - i. Skin: Physical block against microbes.
 - ii. Saliva, tears, mucus: Contain enzymes like lysozyme to kill pathogens.
- c. Internal Defenses:
 - i. Phagocytes: Engulf pathogens.
 - ii. Complement proteins: Enhance immune response.
 - iii. Interferons: Prevent viral replication.
 - iv. Inflammation: Increases blood flow and brings white blood cells to infection.

2. Specific Immunity

- Activated when pathogens get past nonspecific defenses.
- Involves lymphatic system: Lymph nodes, spleen, thymus, etc.
- Lymphocytes:
 - B cells: Produce antibodies that bind to antigens.
 - T cells:
 - Helper T cells: Activate B and cytotoxic T cells.
 - Cytotoxic T cells: Kill infected cells directly.

3. Antibodies & Antigens

- Antigens: Foreign substances that trigger an immune response.
- Antibodies: Proteins made by B cells that bind to specific antigens and help destroy them.

4. Immunity Types

- Active Immunity:
 - Develops after exposure to a pathogen or vaccination.
 - Produces memory cells for long-term protection.
- Passive Immunity:
 - Temporary.
 - Antibodies are received from another source (e.g., breastfeeding).

5. Immune System Failure

- AIDS (Acquired Immunodeficiency Syndrome) is caused by HIV, which attacks helper T cells, weakening the immune response.

